DYP DPU

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Department of Computer Engineering

Cortex: An Agentic Retrieval-Augmented Generation System over Live Work Data with Explainable Memory and Consent-Gated Actions

A Project Presentation for the Degree of Bachelor in Computer Engineering

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Introduction

- **Problem context:** work information is spread across Gmail, Calendar, Drive, Notion, Slack, GitHub and the web, and it changes daily.
- **Classic RAG:** one similarity search over a static corpus; no multi-hop reasoning, no claim checking, silent conflicts.
- **Cortex:** an agentic RAG system that plans sub-questions, searches live sources in parallel and streams an answer in which every factual sentence cites numbered evidence.
- **Trust features:** per-claim verification with a groundedness score, cross-source conflict reporting, explainable memory and consent-gated actions.
- **Cost:** runs entirely on free-tier LLM services through a failover router.

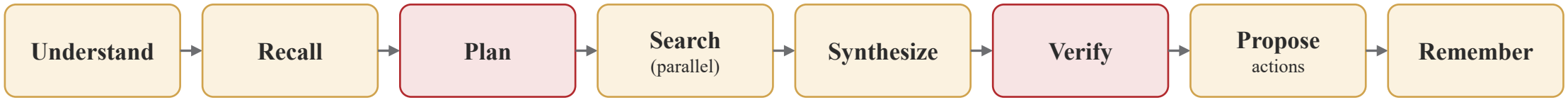


Fig. One bounded agent run: at most 5 sub-questions, 18 tool calls, 4 LLM calls (Deep) and a 90 s timeout.

Literature Survey (1/3)

Ref.	Work	Key contribution	Limitation for live work data
[1]	Lewis et al., RAG (NeurIPS 2020)	Generator conditioned on top-k passages from a dense index	Single retrieval round over a fixed index
[2]	Yao et al., ReAct (ICLR 2023)	Interleaves reasoning traces with tool actions	Open-ended loop; no claim-level check
[3]	Trivedi et al., IRCoT (ACL 2023)	Retrieval interleaved with chain-of-thought for multi-hop QA	Static corpus; many LLM calls
[4]	Press et al., Self-Ask (EMNLP 2023)	Explicit follow-up sub-questions; compositionality gap	No tool or source governance
[5]	Gandhi et al., GRASP (2026) - base paper	RL policy coordinating semantic search, keyword search and paragraph reading	Needs RL training; benchmark corpora only

Literature Survey (2/3)

Ref.	Work	Key contribution	Limitation for live work data
[6]	Asai et al., Self-RAG (ICLR 2024)	Model retrieves on demand and critiques itself with reflection tokens	Requires fine-tuning the generator
[7]	Yan et al., CRAG (2024)	Evaluator scores retrieved documents; falls back to web search	Trained evaluator; web only
[8]	Jeong et al., Adaptive-RAG (NAACL 2024)	Routes questions by complexity	Trained classifier; no live tools
[9]	Gao et al., ALCE (EMNLP 2023)	Benchmark and metrics for citation recall / precision	Evaluation only, not a system
[10]	Min et al., FActScore (EMNLP 2023)	Fraction of atomic facts supported by a source	Offline scoring; costly per fact
[11]	Packer et al., MemGPT (2023)	OS-style paging of agent memory	Memory is opaque to the user

Literature Survey (3/3)

Approach	Multi-step	Training-free	Claim check	Live tools	Consent gate
RAG [1]	No	No	No	No	-
ReAct [2]	Yes	Yes	No	Yes	No
IRCoT [3]	Yes	Yes	No	No	-
Self-RAG [6]	Yes	No	Self-critique	No	-
CRAG [7]	Partly	No	Retrieval check	Web	-
Adaptive-RAG [8]	Adaptive	No	No	No	-
GRASP [5]	Yes	No (RL)	Grounded-reading reward	No	-
Cortex (proposed)	**Yes (bounded)**	**Yes**	**Per claim**	**Yes**	**Yes**

Source: literature survey of the project report (Table: comparison of related approaches).

Gap Finding from Literature

G1. Static corpora

Most RAG methods [1][3][8] retrieve from a fixed, pre-built index, not from live tools that change daily.

G2. Training cost

The strongest retrieval policies (GRASP [5], Self-RAG [6], Search-R1 [12]) need RL or fine-tuning, which free-tier APIs do not allow.

G3. No per-claim audit

Attribution is measured offline [9][10]; deployed assistants rarely show which sentence is supported.

G4. Silent conflicts

When two sources disagree, systems pick one answer instead of reporting the conflict as a risk.

G5. Opaque memory

Agent memory [11][13] is stored on the user's behalf without visible reason, source, expiry or a forget control.

G6. Unsafe actions

Tool-using agents [2][14] can act without per-action approval; MCP widens the attack surface [15].

Motivation

- **Fragmented information costs time:** interaction workers spend about 28% of the week on e-mail and nearly 20% searching for internal information [16].
- **Single-shot RAG is not analysis:** multi-hop questions need several retrieval steps [3][4]; long contexts are used poorly in the middle [17].
- **Trust needs evidence:** attribution can be measured [9][10], so a product can show which claims are supported.
- **Agents that act must ask:** a wrong or injected instruction can send mail or post messages; propose-then-approve is a verifiable safety property.
- **Memory should be explainable:** users should see why something was remembered, where it came from, and be able to forget it.
- **Cost:** students and individual developers cannot pay per-token prices; the design must stay within free tiers.

Problem Statement

To design and implement an agentic retrieval-augmented generation system that answers natural-language questions over a user's live, distributed work data by decomposing each question into sub-questions, retrieving evidence from all relevant sources in parallel within fixed budgets, and producing an answer in which every factual claim is cited and verified against its evidence; that surfaces contradictions between sources, states missing information, keeps an explainable user-governed memory, and performs actions only after explicit per-action consent, while running on free-tier language-model services.

Input
Question (<= 2000 chars), mode (Fast / Deep), recent history

Output
Streamed cited answer, claim states, groundedness G, conflicts, action proposals

Constraint
<= 4 LLM calls (Deep), <= 18 tool calls, 90 s per run, zero cost

Scope & Objective (1/2)

- Answer multi-part questions over several live sources with a bounded plan of sub-questions and parallel retrieval.
- Cite every factual sentence, verify each claim against its evidence and report a groundedness score.
- Surface cross-source contradictions with both citations; state missing information instead of guessing.
- Never execute an action without an explicit, single-use approval of an exact preview.
- Keep long-term memory selective and explainable (reason, source, salience, expiry) with pin, edit and forget.
- Degrade gracefully: name any failed source, explain its impact and continue with the rest.
- Keep LLM calls per question small and predictable (Deep <= 4, Fast <= 2) to stay within free tiers.
- Provide an evaluation harness and automated tests for groundedness, citation recall and consent safety.

Scope & Objective (2/2)

In scope

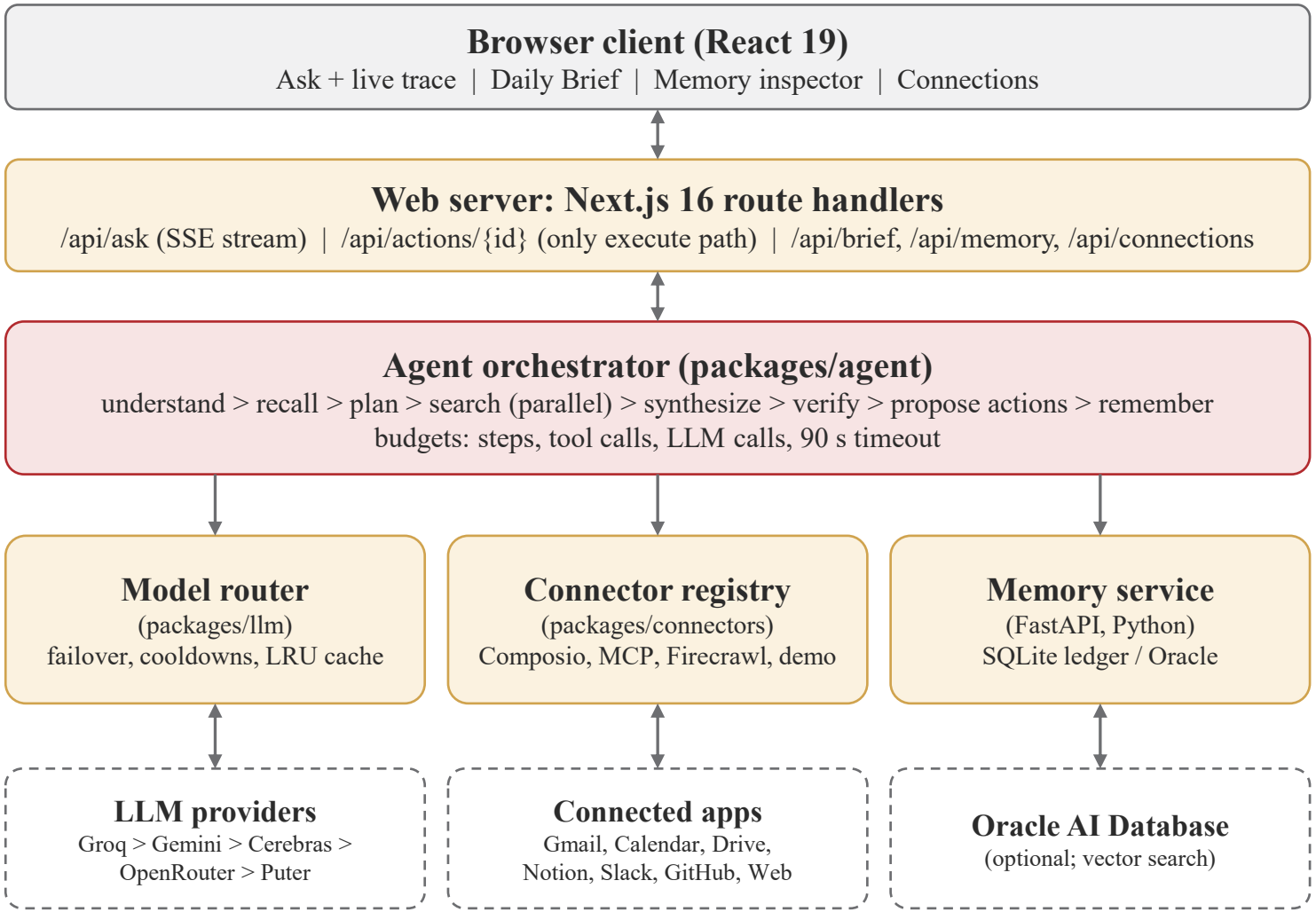
- Web app: Ask with live trace, Daily Brief, Memory inspector, Connections.
- Sources: Gmail, Calendar, Drive, Notion, Slack, GitHub (Composio), any MCP server, web (Firecrawl), sample workspace.
- Memory service with provenance, decay, expiry, pin / edit / forget.
- Consent-gated actions through a single approval endpoint.
- Free-tier LLM router with failover, cooldowns and caching.

Out of scope (this stage)

- Copying users' data into an index or warehouse (tools are queried at question time; 90 s cache only).
- Training or fine-tuning models.
- Autonomous write operations of any kind.
- Time-travel (snapshot diff) queries, editable plan, Cortex-as-an-MCP-server (future work).

Bounds: question <= 2000 chars | history <= 6 turns | <= 5 sub-questions | <= 18 searches | <= 16 evidence items (Deep) | <= 3 proposals and <= 2 memory writes per run

System Architecture



Key design decisions

- Hand-written orchestrator: explicit branches, bounded loops, predictable LLM calls.
- One Connector interface: status, search, execute; new sources by class or MCP config.
- Actions stored server-side as proposals; executed only via approval endpoint (single-use, 30 min).
- Evidence never stored beyond a 90 s search cache.
- Optional LangSmith tracing of runs, LLM calls and searches.

Algorithm / Methods (1/2)

Step	Deep mode	Fast mode	LLM calls
Understand	Check which connectors are live; report the ones that are down	same	0
Recall	Search memory; pinned memories always included	same	0
Plan	Fast-tier model writes 2-5 sub-questions, sources, keyword queries, time window	One sub-question over every source	1 / 0
Search	All (sub-question x source) pairs in parallel, deduplicated, cached 90 s; one broadened retry	Same, no retry	0
Synthesize	Strong model, streamed; answer + CONFLICTS / MISSING / ACTIONS / MEMORY sections	Fast model, shorter	1
Verify	Lexical claim check; LLM re-check only for doubtful claims	Lexical only	0-1 / 0
Actions	Validated against connector schemas, stored, never executed	same	0
Remember	Explicit requests + candidates with salience >= 0.6 (max 2)	same	0

If no evidence is found, Cortex says so without any LLM call. Search latency = max over calls, not the sum (12 s timeout, 1 retry per call).

Algorithm / Methods (2/2)

Evidence ranking	score(e) = 2 h(e,q) + max(0, 3 - a(e)/10) h = question keywords in e; a = age in days (30 if unknown)
Claim verification	o(c) = |T(c) n T(E_c)| / |T(c)| supported: o >= 0.4 and all numbers match; partial: o >= 0.2; else unsupported
Groundedness	G = (1/|L|) Sum w(c), w = 1 / 0.5 / 0 (G = 1 if no claims)
Memory salience decay	s_eff = s x 0.5^(t/30) for unpinned memories; TTL: priority 45 d, fact 90 d
LLM router failover	Groq > Gemini > Cerebras > OpenRouter > Puter; cooldown on 429 (retry-after or 60 s), bad key 10 min, no credit 1 h; 6 h response cache

Requirement Analysis (1/2)

Functional requirements

- **FR1:** ask a question; plan, search and stream a cited answer
- **FR2:** choose Fast or Deep mode
- **FR3:** inspect trace, plan, tool calls and evidence
- **FR4:** approve, edit or decline a proposed action
- **FR5:** read the Daily Brief (changed, at risk, conflicts, upcoming)
- **FR6:** teach, pin, edit or forget memories
- **FR7:** connect an app (OAuth) or use the sample workspace
- **FR8:** add an MCP server through configuration only

Non-functional requirements

- **Performance:** streaming starts within seconds; 12 s per connector call, 25 s per model request, 90 s per run
- **Availability:** provider failover; failed sources skipped and reported; SQLite fallback for memory
- **Security:** single approval path; OAuth tokens never reach the browser; HTTP-only cookie
- **Modifiability:** new source = one class or one MCP config entry
- **Testability:** dependencies injected; 18 automated tests; evaluation harness
- **Usability:** plain-language errors; claims visually marked by verification state

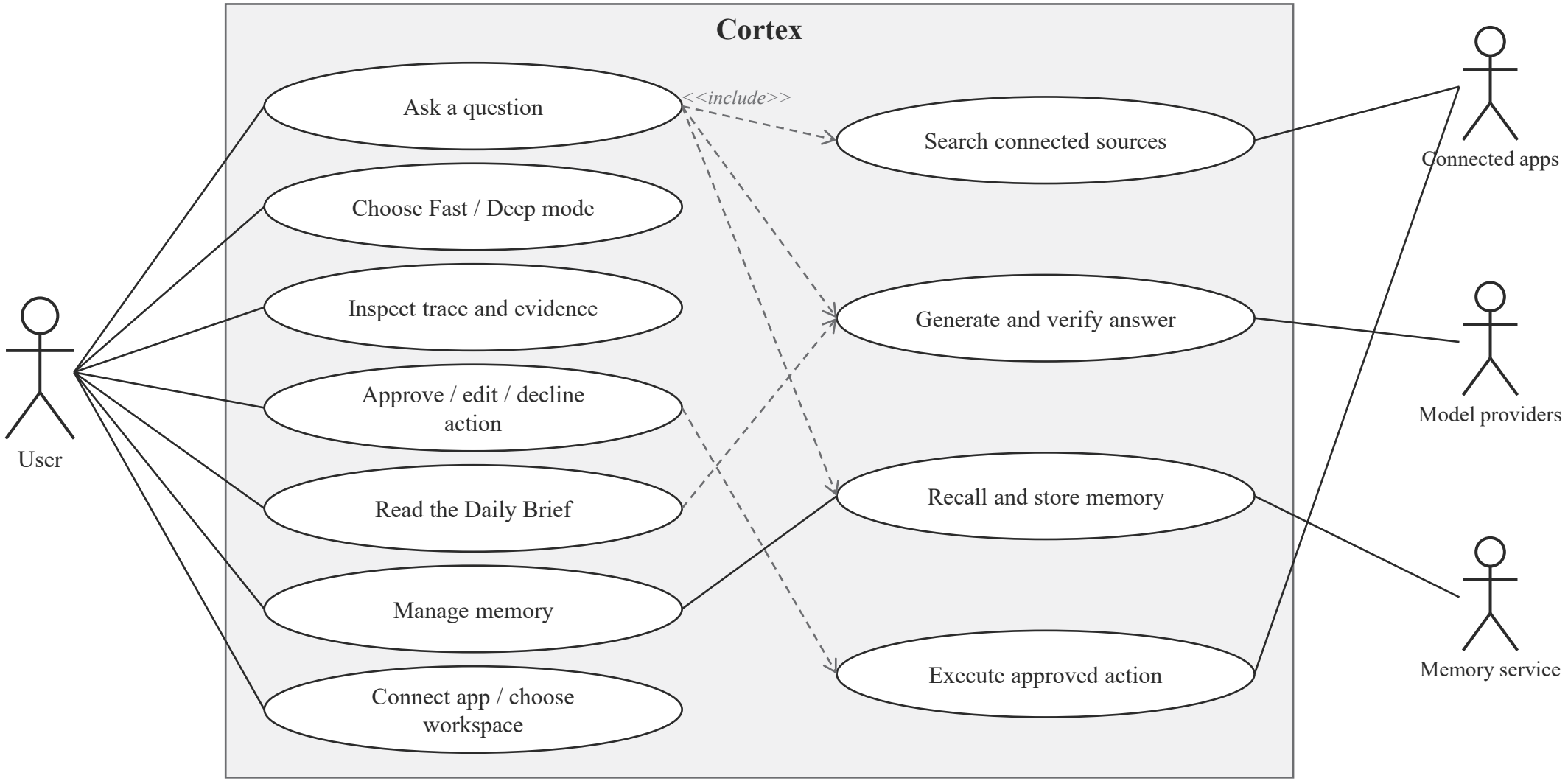
Requirement Analysis (2/2)

Parameter	Minimum requirement
CPU	2 GHz, dual core
RAM	8 GB
Storage	5 GB free
Network	Broadband Internet

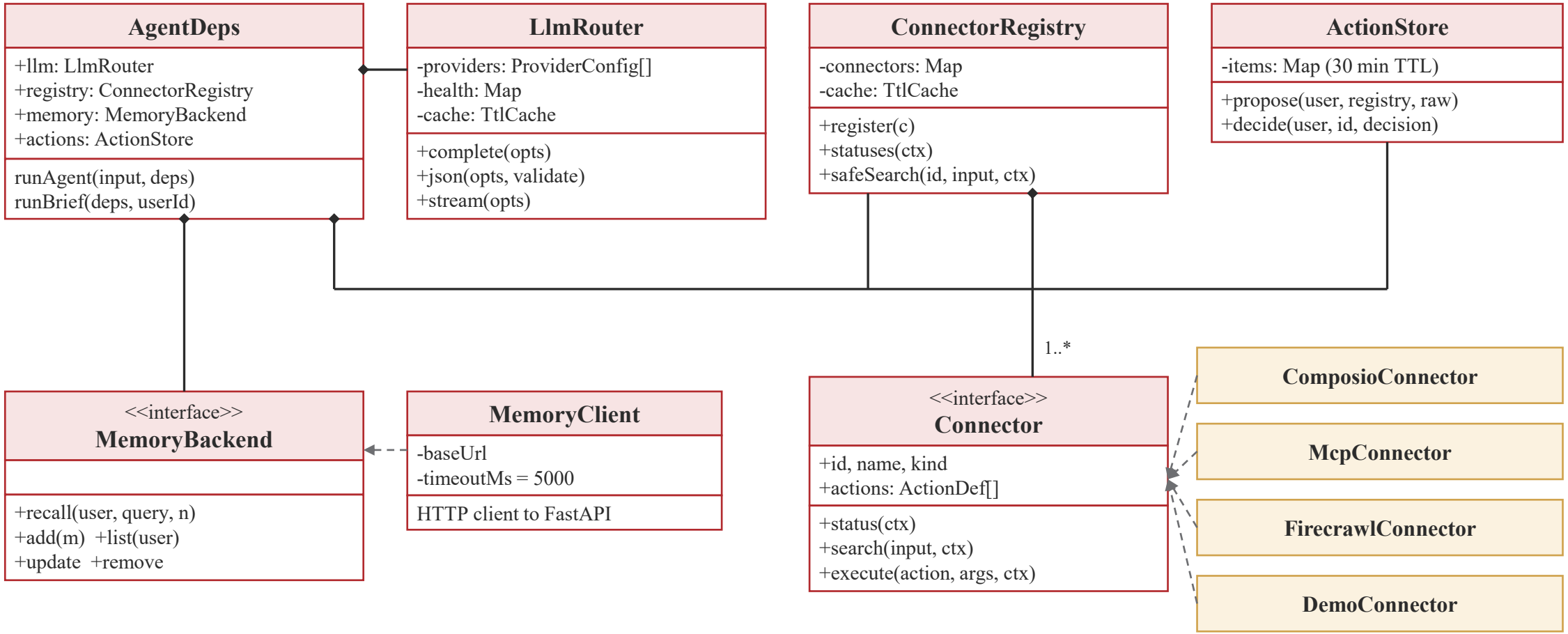
Model inference runs at the providers, so no GPU is needed.

Item	Choice
OS	Windows 10/11, Linux or macOS
Languages	TypeScript (Node.js 24+), Python 3.11
Web	Next.js 16, React 19, Tailwind CSS 4
Agent / tools	Zod, Composio SDK, MCP SDK, Firecrawl
Memory	FastAPI, SQLite; Oracle Agent Memory (optional)
LLMs (free tier)	Groq, Gemini, Cerebras, OpenRouter, Puter
Testing / tracing	Vitest, LangSmith
IDE	Visual Studio Code, Git

UML Diagram: Use Case

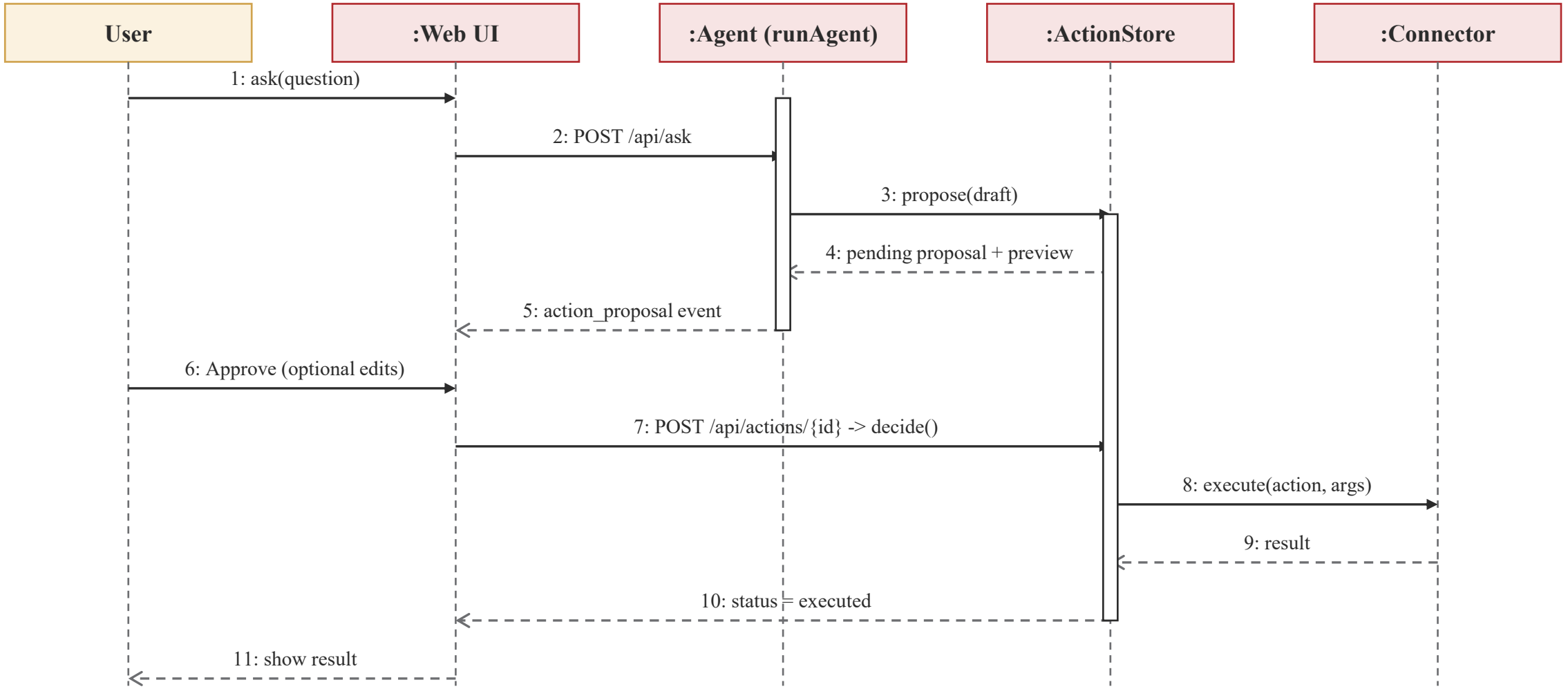


UML Diagram: Class



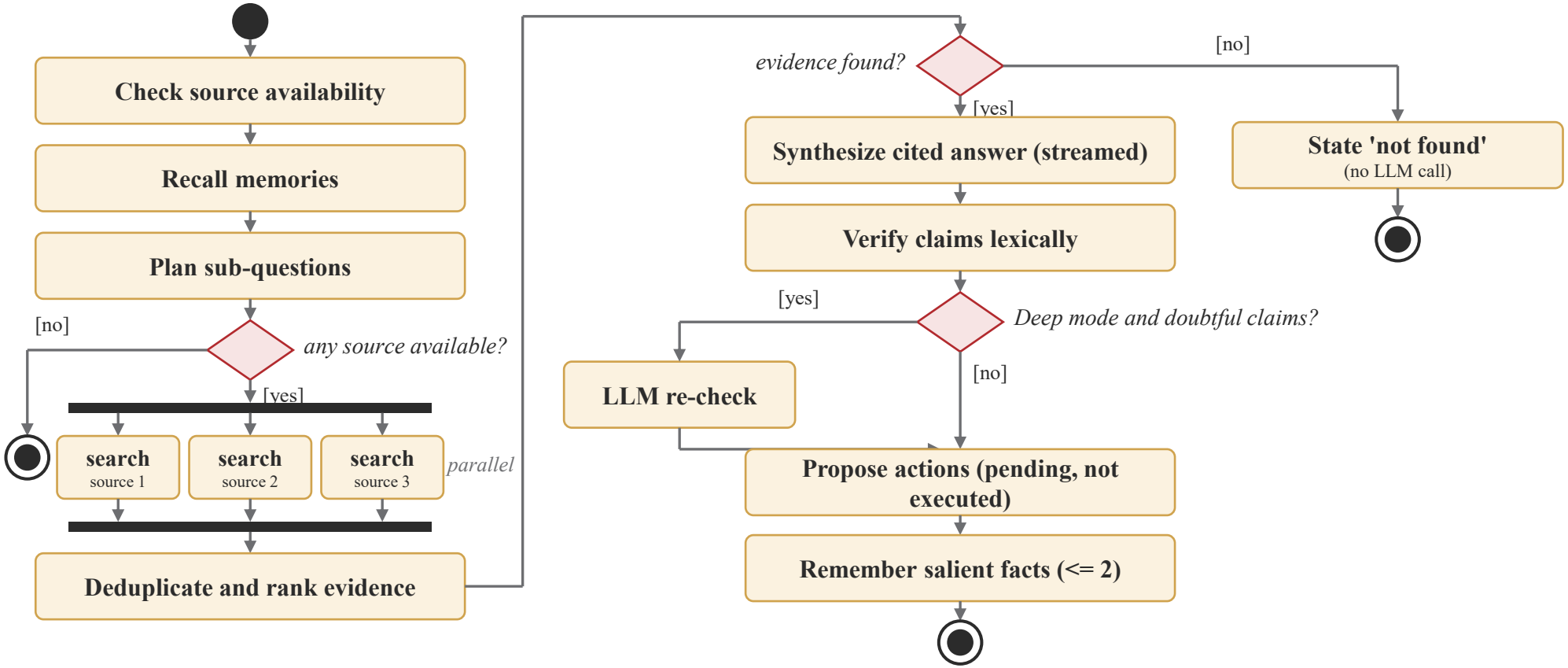
Diamond: AgentDeps / ConnectorRegistry own the linked objects; dashed arrow: realizes an interface. The full two-part class diagram is in the report.

UML Diagram: Sequence



Only step 7 can call connector.execute; it is single-use, scoped to the proposing user and expires after 30 minutes (HTTP 409 on a second decision).

UML Diagram: Activity



Conclusion

- Cortex realises a bounded plan-then-execute agentic RAG pipeline over live work tools, running at zero cost on free-tier LLMs.
- Every factual sentence cites numbered evidence and is verified; conflicts and missing information are reported, not hidden.
- Actions are only proposed; one approval endpoint executes them, once, for the proposing user.
- Memory is selective and explainable, with provenance, decay, expiry, pin, edit and forget.
- Weakest areas so far: conflict detection and action proposals (1 of 2 cases each); deadline questions had the lowest groundedness.

Metric (preliminary)	Value
Mean groundedness	0.88
Citation recall	0.92
Not-found handled	1 / 1
Conflicts detected	1 / 2
Actions proposed	1 / 2
Executed without consent	0
Latency p50 / p90	3.7 s / 7.1 s

Preliminary: one run of 12 developer-written questions on a synthetic workspace, Deep mode, free-tier models (evals/results/deep-2026-10-26.json).

Future Work

- Make conflict detection independent of the synthesis model by comparing dates and numbers extracted from different sources.
- Enlarge the evaluation to the planned 50 questions, repeat runs to report variance, and add citation precision.
- Bring the Oracle AI Database backend into use for vector search over memories and compare it with the SQLite fallback.
- Implement time-travel queries over source snapshots, an editable plan before execution, and shareable reasoning receipts.
- Expose Cortex as an MCP server so that other assistants can use its research and memory.
- Study a learned, granularity-aware retrieval policy in the spirit of GRASP [5] once a trainable model and data are available.
- Deploy to a hosted environment with persistent storage for pending actions.

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Paper Publication

Title: From Static Retrieval to Granularity-Aware Search Policies: A Review of Agentic Retrieval-Augmented Generation with a Case Study on Live Workspace Data

Authors: Ajit Paraskar, Harshvardhan Patil, Prathamesh Kumbhar, Bikram Ghosh (Dept. of Computer Engineering, DIT Pimpri)

Paper details

- Type: review paper with a case study (IEEE conference format)
- Taxonomy of 8 design dimensions of agentic RAG, anchored on GRASP [5]
- Cortex case study with preliminary 12-question evaluation
- 41 references; 2 figures, 4 tables

Status and target venues

- **Status: manuscript prepared; not yet submitted**
- Venue, submission date and decision: to be finalised with the guide
- Candidate venues: SPPU / central university conferences, ACM IKDD CODS-COMAD, IEEE Access, ACL Rolling Review (ACL / EMNLP / NAACL)

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